Gengyi Sun

LinkedIn ORCID Google Scholar
https://sungengyi.github.io/ gengyi.sun@uwaterloo.ca

Publications

International Conference Proceedings

ASE'26 Gengyi Sun, Georges Aaron Randrianaina, Tao Xiao, Yasutaka Kamei, Shane McIntosh. Recovering Revisions of Pull Requests with Altered History *In Proc. of the International Conference on Automated Software Engineering*, To appear, Oct. 2026.

FORGE'26 Evelien Riddell, James Riddell, Gengyi Sun, Michał Antkiewicz, Krzysztof Czarnecki. Reasoning about Multi-hop Fault Propagation with LLM Agents for Root Cause Analysis in Cloud-based Systems. *In Proc. of the International Conference on AI Foundation Models and Software Engineering (FORGE 2026)*, pp. 172-183, Apr. 2026.

ASE'25 Gareema Ranjan, Mahmoud Alfadel, Gengyi Sun, Shane McIntosh. The Cost of Build Tool Downgrades: An Empirical Study of the Kubernetes Project. *In Proc. of the International Conference on Automated Software Engineering (ASE)*, To appear, Nov. 2025.

FSE'24 Gengyi Sun, Sarra Habchi, Shane McIntosh. RavenBuild: Context, Relevance, and Dependency Aware Build Outcome Prediction. *In Proc. of the Int'l Sym. on the Foundations of Software Engineering (ACM)*, pp. 45:1–45:23, Jul. 2024.

ICSE'24 SEIP Gengyi Sun, Mehran Meidani, Sarra Habchi, Mathieu Nayrolles, Shane McIntosh. Code Impact Beyond Disciplinary Boundaries: Constructing a Multidisciplinary Dependency Graph and Analyzing Cross-Boundary Impact. *In Proc. of the Int'l Conf. on Software Engineering, Software Engineering in Practice track (ACM/IEEE)*, pp. 122–133, Apr. 2024.

Short International Conference Papers

ICSE'25 DS Gengyi Sun. Intelligent Automation for Accelerating the Repair of Software Build Failures. *In Proc. of the Int'l Conf. on Software Engineering, Doctoral Symposium track (ACM/IEEE)*, Apr. 2025.

Blogs

Ubisoft Gengyi Sun. Build Outcome Prediction - Saving Time and Resources. Ubisoft, La Forge, Canada. Apr. 2024.
This blog is featured by the official Ubisoft LinkedIn account, which has more than one million followers.

Theses

MMath Gengyi Sun. Quantifying, Characterizing, and Leveraging Cross-Disciplinary Dependencies: Empirical Studies from a Video Game Development Setting. University of Waterloo, Canada. Dec. 2023.

Industrial Experience

Research and Development Intern *(Ubisoft, La Forge)* **Toronto, Canada** *Sept. 2022 - Sept. 2024*
Researched the Continuous Integration (CI) process in the context of video games.

Dependency Graph Analytics

Formulated empirical analyses to identify changes of higher risks based on the dependency graphs of the video game project.

Build Outcome Prediction

Proposed strategies to reduce CI costs and improve CI stability. Multiple teams at Ubisoft have invested in developing API services to enable internal feedback cycles for this project. *This work was also presented at Ubisoft Developer Conference 2024 and Game Developer Conference 2024, attracting interests from within and outside of the company.*

Committee Service

MSR'27 *PC for Technical Papers* *2027*
EAAI *Journal Reviewer* *2026*
TOSEM *Journal Reviewer* *2026*
AIware'26 *Publicity Co-Chair* *2026*
MSR'26 *PC for Mining Challenge* *2026*
ICSE'26 *Shadow Research Track PC* *2026*
MSR'25, 26 *Junior PC for Technical Papers* *2025 - 2026*

Awards

David R. Cheriton Graduate Scholarships *$20,000 total, 2024 - 2025* *Sept. 2025*
David Johnston International Entrance Scholarship *$5,000* *Apr. 2025*
Arthur Levine SURE Award *$2,800* *May. 2020*

Academic Collaboration

Visiting Researcher *POSL Research Group, Kyushu University* **Fukuoka, Japan** *Feb. - Apr. 2025*
Collaborated with Prof. Yasutaka Kamei on research related to automatically addressing software build failures.

Invited Presentations

Taming the beast: use ML to curb CI costs *at UDC 2024* **Montréal, Canada** *Feb. 2024*
Ubisoft Developer Conference (UDC) is an Ubisoft internal application-based conference. This talk is broadcast online to all Ubisoft employees globally (800 onsite viewers and 3,469 online viewers).

Activities

Computer Science Departmental Director *MathGSA, University or Waterloo* *Sept. 2025 - present*
President *Computer Science Graduate Student Association, University or Waterloo* *Jul. 2024 - Aug. 2025*
Student Volunteer *ASE'25, ICSE'25, FSE'24,26* *May. 2025*
Ubisoft Ambassadors *Hack-The-North 2023* *Sept. 2023*

Education

Ph.D, Computer Science *University of Waterloo* **Waterloo, Canada** *May 2024 - present*
Supervisor: Prof. Shane McIntosh

MMath, Computer Science *University of Waterloo* **Waterloo, Canada** *Sept. 2021 - Dec. 2023*
Supervisor: Prof. Shane McIntosh

B. Software Eng *McGill University* **Montréal, Canada** *Sept. 2017 - Apr. 2021*